



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
STA. MESA MANILA
COLLEGE OF ENGINEERING

Self-Study Report (SSR)

Bachelor of Science in Civil Engineering

Program Specific Requirements

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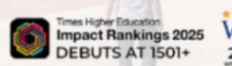
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II. SPECIFIC PROGRAM CRITERIA

Describe how the program satisfies any applicable specific program criteria. If already covered elsewhere in the self-study report, provide appropriate references

1. Introduction to the Specific Program Criteria

The Bachelor of Science in Civil Engineering (BSCE) program at the Polytechnic University of the Philippines is designed to produce competent, innovative, and socially responsible Civil Engineering professionals equipped with a strong foundation in mathematics, natural sciences, engineering sciences, civil engineering principles, and emerging technologies.

The program adheres to the outcomes-based education (OBE) framework and aligns with the standards prescribed by the Commission on Higher Education through *CMO No. 92, series of 2017, otherwise known as the "Policies, Standards, and Guidelines for the Bachelor of Science in Civil Engineering (BSCE) Program."* The curriculum further supports the accreditation requirements of the Philippine Technological Council under the Washington Accord framework by ensuring that graduates attain the necessary knowledge, skills, attitudes, and professional competencies expected of globally competitive Civil Engineering practitioners.

In terms of breadth, the program provides students with comprehensive exposure to general education, mathematics, physical sciences, computing fundamentals, engineering sciences, and allied disciplines necessary for multidisciplinary understanding and lifelong learning. The curriculum incorporates foundational and professional courses covering structural engineering, geotechnical engineering, construction management engineering, transportation engineering and water resources engineering. These curricular components enable students to develop a broad understanding of contemporary technologies, engineering practice, ethical responsibilities, environmental awareness, and societal impact.

In terms of depth, the program ensures intensive specialization and advanced application of Civil Engineering principles through laboratory experimentation, design courses, project-based learning, research activities, internships, and capstone projects. Students are provided opportunities to engage in complex engineering problem-solving and innovation-driven projects aligned with industry and national development needs. The program

emphasizes analytical thinking, technical proficiency, design competence, and the application of modern engineering tools necessary for professional practice and advanced studies.

The BSCE program continuously undergoes curriculum review, assessment, and Continuous Quality Improvement (CQI) processes to ensure full compliance with CMO No. 92 s. 2017, and alignment with the graduate attributes and program outcomes required by the Philippine Technological Council. Through these mechanisms, the program maintains academic excellence, industry relevance, and international comparability consistent with the standards of engineering education under the Washington Accord.

Table No. 1 – *Civil Engineering Specializations demonstrating depth aligned with both foundational (basic) and advanced competencies.*

No.	SPECIALIZATION	SPECIALIZED SUBJECTS
1	Construction Engineering and Management	• CMEN 451 Project Construction and Management • CMEN 452 Construction Planning & Scheduling • CMEN 461 Advanced Construction Methods and Equipment • CMEN 462 Construction Productivity • CMEN 471 Construction Cost Engineering • CMEN 472 Construction Procurement • CMEN 481 Database Management in Construction • CMEN 482 Construction Project Administration • CMEN 491 Construction Occupational Safety and Health (COSH) • CMEN 492 Total Quality Management for Civil Engineering
2	Geotechnical Engineering	• GTEN 451 Geosynthetics in Geotechnical Engineering • GTEN 452 Geotechnical and Geo-environmental Engineering • GTEN 461 Geotechnical Earthquake Engineering • GTEN 462 Underground Infrastructure Engineering • GTEN 471 Geotechnical Aspects of Landfill Design • GTEN 472 Slope Stability and Earthworks

		• GTEN 481 Geotechnical Engineering (Rock Mechanics) • GTEN 482 Soil & Rock Dynamics • GTEN 491 Ground Improvement • GTEN 49 Environmental Geotechnology
3	Structural Engineering	• STEN 452 Finite Analysis • STEN 461 Prestressed Concrete Design • STEN 462 Wind Engineering • STEN 471 Dam Engineering • STEN 472 Design of Steel Structures • STEN 481 Bridge Engineering • STEN 482 Foundation and Retaining Wall Design • STEN 491 Structural Design of Towers and Other Vertical Structures • STEN 492 Structural Dynamics
4	Transportation Engineering	• TREN 451 Transportation Systems Planning and Design • TREN 452 Traffic Engineering • TREN 461 Highway Engineering • TREN 462 Pavement Analysis and Design • TREN 471 Railroad Engineering • TREN 472 Project Evaluation & Appraisal of Transportation Facilities • TREN 481 Airport Design • TREN 482 Urban and Regional Transportation Systems Planning • TREN 491 Ports and Harbors • TREN 492 Transportation Modeling
5	Water Resources Engineering	• WREN 451 Water Resources Engineering • WREN 452 Multipurpose Water Resources Development • WREN 461 Flood Control and Drainage Design • WREN 462 Hydraulic Design • WREN 471 Irrigation Engineering • WREN 472 Sanitary and Wastewater Engineering

		• WREN 481 Water Supply Planning and Development • WREN 482 Groundwater Flow Modeling • WREN 491 River Engineering • WREN 492 Coastal Engineering
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2. Alignment of Curriculum with Specific Program Criteria

Discuss how the curriculum addresses the required knowledge areas.

The Bachelor of Science in Civil Engineering (BSCE) program is designed to systematically address the required knowledge areas of Civil Engineering through a well-structured curriculum that integrates both lecture-based instruction and laboratory-based experiential learning. This design ensures compliance with CMO No. 92 s. 2017 and alignment with the program outcomes and graduate attributes prescribed by the Philippine Technological Council under the Washington Accord framework.

The curriculum is organized to provide a balanced development of foundational knowledge, engineering sciences, and specialized Civil Engineering competencies. Lecture components deliver the theoretical underpinnings necessary for understanding core concepts in civil engineering. These lecture-based courses emphasize analytical thinking, problem-solving, conceptual understanding, and the application of civil engineering principles.

Complementing the lecture components, laboratory courses and integrated laboratory activities reinforce theoretical concepts through hands-on experimentation, simulation, and system implementation. Laboratory work enables students to apply engineering knowledge in practical settings using civil engineering modern tools and software platforms.

The curriculum ensures coverage of key knowledge areas in Civil Engineering such as structural engineering, geotechnical engineering, construction management engineering, transportation engineering and water resources engineering. Each of these areas is addressed through a combination of lecture discussions and laboratory exercises, ensuring that students not only understand theoretical principles but also acquire the ability to implement and validate engineering solutions.

Lecture components are designed to build cognitive understanding of engineering theories, models, and analytical methods, while laboratory components provide experiential learning that develops psychomotor and applied competencies. This dual approach strengthens students' ability to bridge theory and practice, particularly in system design, experimentation, and problem-solving in real-world engineering contexts.

Furthermore, laboratory activities are aligned with course outcomes and are assessed using structured rubrics that measure technical skills, design capability, analytical reasoning,

teamwork, and documentation. Capstone design integration, simulation exercises, and project-based learning further enhance the depth of learning by requiring students to synthesize knowledge from multiple courses and apply it to complex engineering problems.

Through the integrated implementation of lecture and laboratory components, the BSCE curriculum ensures that students achieve both breadth and depth of knowledge in Civil Engineering. This approach produces graduates who are capable of applying engineering principles effectively in analysis, design, implementation, and evaluation of computing systems, consistent with national standards and international engineering education benchmarks.

References:

All references and appendices on the specific program criteria narrative are all incorporated in the Criterion 5 in the self-study report.

3. Mapping of Specific Criteria to Courses

Explain that the curriculum mapping ensures coverage of competencies.

The Bachelor of Science in Civil Engineering (BSCE) program implements a comprehensive curriculum mapping system to ensure that all required competencies prescribed under CMO No. 92 s. 2017 and the graduate attributes required by the Philippine Technological Council are adequately addressed, developed, and assessed throughout the curriculum. The curriculum mapping process serves as a fundamental mechanism in ensuring constructive alignment among Program Educational Objectives (PEOs), Student Outcomes (POs), Course Outcomes (COs), Teaching-Learning Activities (TLA's), and Assessment Task (AT's) consistent with the principles of Outcomes-Based Education (OBE).

The curriculum mapping ensures that the breadth and depth of Civil Engineering knowledge and competencies are systematically distributed across general education, mathematics, natural sciences, engineering sciences, professional courses, and design-oriented experiences. Each course in the curriculum is mapped to specific Student Outcomes to identify its contribution to the development of technical knowledge, engineering problem-solving abilities, design competencies, ethical responsibility, communication skills, teamwork, lifelong learning, and understanding of societal and environmental impacts.

The program utilizes a CO - SO mapping matrix to determine the level of contribution of each course outcome to the attainment of Student Outcomes. Foundational courses provide introductory competencies in mathematics, sciences, computing, and engineering fundamentals, while professional courses progressively develop higher-level competencies in civil engineering. Advanced courses, capstone projects, internships, and research activities serve as culminating experiences that enable students to integrate and apply acquired knowledge and skills in solving complex engineering problems.

The mapping process also ensures vertical and horizontal alignment of learning competencies throughout the curriculum. Vertical alignment guarantees the progressive development of competencies from introductory to advanced levels, while horizontal alignment ensures integration and reinforcement of related competencies across courses within the same academic level. This systematic alignment prevents unnecessary duplication of topics and ensures adequate coverage of all required knowledge areas in Civil Engineering.

To ensure the effectiveness of curriculum implementation, the program regularly conducts curriculum reviews and validation activities involving faculty members, industry practitioners, alumni, students, and other stakeholders. The results of assessments, performance indicators, licensure examinations, internship evaluations, capstone outputs, employer feedback, and outcomes assessment are analyzed to determine the extent of Student Outcome attainment and identify areas for improvement. These findings are utilized in the Continuous Quality Improvement (CQI) process to enhance course content, instructional strategies, assessment methods, laboratory resources, and curriculum structure.

Furthermore, curriculum mapping supports the identification of competencies related to complex engineering problem-solving, modern tool usage, experimentation, design and development of solutions, project management, ethics, sustainability, and lifelong learning as required under the Washington Accord framework. Performance indicators and assessment tools are established to measure student achievement at various stages of the program, ensuring that graduates possess the expected competencies of a Civil Engineering professional.

Through its systematic curriculum mapping and continuous assessment mechanisms, the BSCE program ensures that graduates acquire the necessary knowledge, skills, values, and professional attributes required for successful engineering practice, advanced studies, research, innovation, and contribution to national and global technological development.

References:

All references and appendices on the specific program criteria narrative are all incorporated in the Criterion 5 in the self-study report.

4. Laboratory and Facilities Support

Discuss laboratories and specialized facilities.

The Bachelor of Science in Civil Engineering (BSCE) program maintains adequate laboratories and specialized facilities that support the attainment of the Student Outcomes and the Specific Program Criteria prescribed by the Philippine Technological Council under the Washington Accord framework. These facilities provide students with opportunities for experiential learning, experimentation, system design, simulation, troubleshooting, research, and project development consistent with the requirements of CMO No. 92 s. 2017 and current industry practices.

The program laboratories are designed to reinforce theoretical concepts through practical applications and hands-on activities essential in the field of Civil Engineering. The facilities enable students to acquire competencies in drafting, surveying, hydraulics, construction materials and testing, geotechnical engineering and computer simulation. Laboratory experiences are integrated into professional courses to ensure that students develop technical proficiency, analytical thinking, problem-solving skills, and the ability to use modern engineering tools necessary for engineering practice.

The program is supported by *specialized laboratories such as the drafting*, surveying, hydraulics, construction materials and testing, geotechnical engineering and computer simulation, and development tools necessary for conducting experiments and implementing engineering projects.

The laboratories and specialized facilities are regularly evaluated, upgraded, and maintained to ensure adequacy, functionality, safety, and relevance to current technological trends and industry requirements. Laboratory manuals, safety procedures, utilization schedules, and maintenance programs are implemented to support effective instructional delivery and responsible use of resources. Faculty members and laboratory personnel provide technical supervision and assistance to ensure that students maximize learning opportunities and adhere to proper engineering and safety practices.

Furthermore, the facilities support the conduct of undergraduate research, design projects, technical training, competitions, internships, and extension activities that strengthen students' practical competencies and professional preparation. Capstone projects and experimental studies conducted in these laboratories allow students to apply engineering

principles, modern engineering tools, and design methodologies in addressing real-world engineering problems and societal needs.

Through these laboratories and specialized facilities, the BSCE program ensures that students are provided with a conducive learning environment that promotes innovation, technical excellence, and experiential learning aligned with industry expectations, national standards, and the accreditation requirements of the Philippine Technological Council.

References:

All references and appendices on the specific program criteria narrative are all incorporated in the Criterion 6 and 7 in the self-study report.

5. Faculty Qualifications

Explain faculty expertise related to the discipline.

The Bachelor of Science in Civil Engineering (BSCE) program is supported by a pool of qualified, competent, and professionally engaged faculty members whose expertise is aligned with the disciplinary requirements of Civil Engineering as per CMO 92 s.2017 and the standards prescribed by the Philippine Technological Council under the Washington Accord framework. The faculty members possess the academic qualifications, technical competencies, industry exposure, research capability, and professional engagement necessary to effectively deliver the curriculum and support the attainment of Student Outcomes.

The program faculty consists of licensed civil engineers, aligned graduate degree holders, researchers, industry practitioners, and specialists in various fields of Civil Engineering and allied disciplines. Their diverse academic and professional backgrounds contribute to the breadth and depth of instruction and ensure that students are exposed to current engineering practices, emerging technologies, and industry trends.

Faculty members handling professional Civil Engineering courses possess relevant educational qualifications at the master's and doctoral levels in Civil Engineering and related fields. These qualifications strengthen the faculty's capability to deliver advanced theoretical instruction, facilitate laboratory activities, supervise research projects, and mentor students in technical and design-oriented courses.

In addition to academic preparation, faculty members maintain active involvement in professional organizations, research activities, extension programs, industry collaborations, and continuing professional development initiatives. Participation in seminars, workshops, conferences, certifications, and technical training programs enables faculty members to continuously update their knowledge and competencies in alignment with rapidly evolving technologies and engineering standards. Such engagements ensure that instructional delivery remains responsive to industry needs and technological advancements.

The program also benefits from faculty members with relevant industry experience and professional practice in the different fields of civil engineering. Their practical experiences enrich classroom instruction through the integration of real-world engineering applications, case studies, design methodologies, and industry-based problem-solving approaches.

Faculty expertise further supports the implementation of Outcomes-Based Education (OBE), assessment of Program Outcomes, and Continuous Quality Improvement (CQI) processes of the program. Faculty members are actively involved in curriculum development, curriculum mapping, outcomes assessment, preparation of instructional materials, laboratory development, and evaluation of student performance. Their participation in academic planning and quality assurance initiatives ensures that the program remains compliant with CMO No. 92 s. 2017 and aligned with the accreditation requirements of the Philippine Technological Council.

Research productivity and scholarly engagement among faculty members also contribute significantly to the advancement of the program. Faculty members undertake research and innovation activities in areas relevant to Civil Engineering, resulting in research presentations, publications and collaborative projects with academic institutions, government agencies, and industry partners. These scholarly activities promote a culture of research, innovation, and continuous learning among both faculty and students. Furthermore, faculty members serve as advisers and mentors for undergraduate research, capstone projects, student organizations, technical competitions, and community extension initiatives. Through these roles, they guide students in applying engineering principles, conducting investigations, developing innovative solutions, and addressing real-world technological challenges consistent with professional and ethical standards.

The program continuously evaluates faculty performance through established institutional mechanisms covering instruction, research, extension, professional development, and community engagement. Results of faculty evaluations, stakeholder feedback, and professional development assessments are utilized to identify training needs and improve instructional effectiveness and faculty competencies. Through its qualified and professionally competent faculty members, the BSCE program ensures effective delivery of the curriculum, achievement of Student Outcomes, enhancement of research and innovation, and preparation of graduates who are globally competitive, ethically responsible, and capable of addressing complex engineering problems in the field of Civil Engineering.

References:

All references and appendices on the specific program criteria narrative are all incorporated in the Criterion 4 in the self-study report.

6. Capstone, Research, and Industry Exposure

Discuss how students apply engineering principles.

The Bachelor of Science in Civil Engineering (BSCE) program provides structured opportunities for students to apply engineering principles through capstone design projects, research activities, and industry exposure, in alignment with the program outcomes and graduate attributes prescribed by the Philippine Technological Council under the Washington Accord framework and the requirements of CMO No. 92 s. 2017.

Students undertake capstone design projects and research activities that address real-world engineering problems. These projects integrate civil engineering solutions and are evaluated based on technical design, innovation, sustainability, ethics, and societal impact. In the design and implementation process, students are required to apply core engineering principles in the analysis.

Research activities further strengthen the application of engineering principles by engaging students in systematic investigation, data analysis, modeling, simulation, and interpretation of engineering problems. Students apply scientific and computational methods to generate evidence-based conclusions and propose innovative solutions. Research topics commonly address the different civil engineering specializations.

Internship and industry immersion programs further expose students to professional engineering practice and emerging technologies. During industry deployment, students apply theoretical knowledge and technical skills in actual workplace environments. These experiences allow students to understand industry standards, engineering workflows, project management practices, quality assurance processes, and multidisciplinary collaboration.

Industry exposure also reinforces students' ability to operate under real engineering constraints such as production deadlines, resource limitations, client requirements, regulatory standards, and system reliability expectations. Under the supervision of industry practitioners and faculty coordinators, students develop professional competencies in communication, teamwork, adaptability, and ethical responsibility while applying engineering principles to solve practical problems.

Collectively, the integration of capstone design projects, research activities, and industry immersion ensures that students of the BSCE program develop the ability to apply engineering

principles in solving complex problems involving multiple constraints, informed trade-off decisions, and real-world applications. This comprehensive approach demonstrates full alignment with the outcomes-based standards of the Philippine Technological Council and ensures that graduates are prepared for professional engineering practice, innovation, and lifelong learning in the field of Civil Engineering.

References:

All references and appendices on the specific program criteria narrative are all incorporated in the Criterion 3, 4, 5, and 6 in the self-study report.

7. Continuous Quality Improvement (CQI)

Show how the program improves based on the assessment.

The Bachelor of Science in Civil Engineering (BSCE) program demonstrates a systematic and sustained commitment to Continuous Quality Improvement (CQI) in alignment with the outcomes-based education framework, the standards of CMO No. 92 s. 2017, and the accreditation requirements of the Philippine Technological Council under the Washington Accord.

The program implements CQI mechanisms through periodic assessment of Student Outcomes (SOs), structured stakeholder feedback, curriculum review activities, industry consultations, and analysis of licensure examination results and graduate employment performance. These assessment mechanisms ensure that the program remains responsive to evolving industry demands, technological advancements, and professional engineering standards in the field of Civil Engineering.

Student Outcome assessment is conducted through direct and indirect measurement tools such as course outcome evaluations, performance rubrics, examinations, laboratory outputs, capstone design assessments, internship evaluations, and student portfolios. The results of these assessments are systematically analyzed to determine the level of attainment of program outcomes and identify areas requiring improvement. Where gaps are identified, appropriate corrective and preventive actions are implemented at the course and program levels.

Stakeholder feedback plays a critical role in program enhancement. Inputs from students, alumni, industry partners, faculty members, and external evaluators are gathered through surveys, consultations, focus group discussions, and advisory meetings. These inputs provide valuable insights on curriculum relevance, instructional effectiveness, laboratory adequacy, graduate competencies, and emerging industry requirements. The feedback is carefully analyzed and integrated into curriculum enhancement plans and instructional improvements.

Curriculum review and mapping activities are regularly conducted to ensure alignment with Student Outcomes, industry standards, and Washington Accord graduate attributes. Through these reviews, course contents are updated to incorporate the civil engineering specializations. Redundant topics are refined, prerequisite structures are improved, and new

courses or modules are introduced as needed to strengthen both breadth and depth of Civil Engineering education.

Industry consultation further strengthens CQI implementation by ensuring that the program remains relevant to current engineering practice. Industry partners provide recommendations on skill requirements, technological trends, and workplace competencies expected from graduates. These inputs are used to enhance capstone project directions, internship experiences, laboratory exercises, and specialized course content, thereby improving graduate employability and professional readiness.

Analysis of licensure examination performance and graduate employment data provides quantitative evidence of program effectiveness. Trends in board examination results and employment rates are evaluated to assess the adequacy of instructional strategies and curriculum design. Areas of weakness identified through these analyses are addressed through targeted interventions such as faculty development, curriculum revision, enhanced laboratory activities, and student academic support programs.

The results of all CQI-related assessments are systematically documented and translated into actionable improvement plans. These improvements are reflected in the enhancement of instructional delivery methods, upgrading of laboratory facilities and equipment, refinement of curriculum content, strengthening of assessment tools, and improvement of student support services such as tutoring, mentoring, and academic advising.

Through this structured CQI system, the BSCE program ensures continuous alignment with industry needs, academic standards, and accreditation requirements. The process reinforces a culture of evidence-based decision-making, accountability, and continuous improvement, ultimately ensuring that graduates possess the competencies, professionalism, and adaptability required in the dynamic field of Civil Engineering.

References:

All references and appendices on the specific program criteria narrative are all incorporated in the Criterion 9 in the self-study report.